EEBUS SPEAK ENERGY

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EEBus UC Implementation Guideline

General Guidelines

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1 Scope of the document

This document provides general, cross-use-case implementation guidelines for the EEBUS architecture (short-name for this document: IG-General).

While specific Use Case Implementation Guidelines (e.g., for Limitation of Power Consumption) address the unique choreography and data requirements of individual scenarios, this document details how to interpret overarching concepts, data structures and behaviors that apply to multiple or all EEBUS Use Cases.

It provides examples and best practices to prevent common implementation errors that frequently cause interoperability issues or certification failures.

Furthermore, this guideline may contain specific modifications that intentionally deviate from the current general Use Case specifications for practical or technical reasons. These retroactive changes are additionally covered by separate errata sheets, whereas the general clarifications are exclusively documented in this Implementation Guide.

This document will be continuously updated as new cross-use-case issues requiring clarification or correction are identified

1.1 References

1.1.1 EEBUS documents

[SPINE] *Minimum*: EEBus_SPINE_V1.3.0_Final.zip

[SHIP] *Minimum*: EEBus_SHIP_TS_Specification_v1.0.1.pdf
Recommended: EEBus_SHIP_TS_Specification_v1.1.0.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels. Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor
An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

CEM
Abbreviation for "Customer Energy Manager". The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimised operation of the connected devices by harmonising energy demand and availability.

Controllable System
Actor name for a logical or physical appliance or system that is remote controllable (e.g. via power limits).

Energy Guard
Actor name for a logical or physical unit capable of managing the energy demand of connected devices.

EVSE
Abbreviation for "Electric Vehicle Supply Equipment"

IG-General
Abbreviation for "Implementation Guideline for Use Case Limitation of Power Consumption" (short name of this document)

LPC
Abbreviation for the Use Case "Limitation of Power Consumption"

MPC
Abbreviation for the Use Case "Monitoring of Power Consumption"

RFE
Abbreviation for "Restricted Function Exchange"

Scenario
Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

SPINE
Abbreviation for "Smart Premises Interoperable Neutral-message Exchange": Technical Specification of EEBus Initiative e.V.

UC
Abbreviation for "Use Case"

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Notes on requirements in this document

This document clarifies overarching requirements that apply to EEBUS Use Cases.

Regarding the core specifications, the keywords "SHALL" and "SHALL NOT" are generally already covered by mandatory requirements of the respective Use Case or SPINE specifications. However, in cases where this guideline explicitly defines retroactive changes to overarching rules, these keywords express mandatory requirements that intentionally override the original specifications. Such modifications are additionally tracked via errata sheets.

2 General clarifications on the high-level part of EEBUS Use Cases

2.1 Definition of "Client Actors" and "Server Actors"

Within the EEBUS architecture, it is essential to distinguish between the protocol layer (SPINE) and the application layer (Use Cases). On the SPINE protocol layer, the roles Client and Server are strictly defined for individual Features (e.g., a Server Feature provides data and accepts writes, while a Client Feature requests or modifies that data).

On the Use Case layer, functions are organized via high-level constructs called Actors (e.g., EnergyGuard, Controllable System, MonitoringAppliance, MonitoredUnit). While a SPINE Entity can support multiple Use Cases simultaneously and assume different Actor roles for each, the overall choreography within a specific Use Case typically follows a directional master-slave or controller-device relationship.

To establish a clear link between these high-level Actor roles and the underlying SPINE communication mechanics, the Actor roles within any Use Case are categorized into Client Actors and Server Actors based on the following functional criteria:

2.1.1 Client Actor

The Client Actor is the orchestrating, controlling or data-collecting SPINE Entity within a specific Use Case. It represents the primary consumer of the Use Case's core functionality.

- Functional characteristics:
    - It typically initiates the functional choreography by actively requesting data or writing configuration/limit values to the remote SPINE Entity.
    - Its primary purpose is to monitor, manage or optimize one or multiple connected systems.
    - Examples: The "EnergyGuard" in the "Limitation of Power Consumption" (LPC) Use Case or the "MonitoringAppliance" in the "Monitoring of Power Consumption" (MPC) Use Case.
- Protocol mapping: The Client Actor predominantly implements the SPINE Client Features required to interact with the core data models of the Use Case.

2.1.2 Server Actor

The Server Actor is the executing, managed or data-providing SPINE Entity within a specific Use Case. It represents the provider of the resource or information.

- Functional characteristics:
    - It exposes its status, measurements or capabilities to the network and reacts to incoming commands or limitations.
    - It executes the core physical or logical behavior demanded by the Use Case (e.g., throttling power consumption or providing meter readings).
    - Examples: The "ControllableSystem" in the "Limitation of Power Consumption" (LPC) Use Case, or the "MonitoredUnit" in the "Monitoring of Power Consumption" (MPC) Use Case.

- Protocol mapping: The Server Actor predominantly implements the SPINE Server Features that hold the state and resource data relevant to the Use Case.

2.1.3 Handling secondary functions (e.g., heartbeats)

A Use Case may include secondary or auxiliary functions where the typical communication direction is reversed. For instance, in the LPC Use Case, the "EnergyGuard" (Client Actor) hosts a Server Feature to provide its own heartbeat to the "ControllableSystem" (Server Actor).

- Rule of interpretation: The categorization as a Client or Server Actor is determined strictly by the primary focus and core functionality of the Use Case (e.g., the act of limiting power in LPC or collecting measurements in MPC). Secondary sub-functionalities, such as cross-checking connection states via heartbeats, SHALL NOT invert or alter the general Client/Server classification of the respective Actor roles.

Note: A complete mapping of all Use Cases to their respective Client and Server Actor classifications can be found in Annex A of this document.

2.2 Multiple energy management devices in one setup

Care must be taken when multiple energy management devices are active in the same setup (local network).

Currently, there is no mechanism that ensures a deterministic behaviour of the managing devices and the managed devices. The only way to run a setup like this without any problems is to connect only ONE manager per managed device. As this cannot be easily guaranteed within the business logic of the devices, a clear warning should be provided in the user manuals.

3 General clarifications on the technical part of EEBUS Use Cases

3.1 Adherence to Use Case specific mandatory data

An architectural principle of EEBUS is the separation between the generic data model (SPINE Resource Specification) and the application-specific requirements (Use Case specifications).

To enable mechanisms like Restricted Function Exchange (RFE), the underlying XML schema definitions (XSD) of SPINE define almost all elements as optional (minOccurs="0"). However, this structural optionality does NOT imply that these elements can be omitted in a specific application context.

Requirement:

- For a server: If a Use Case specification defines an Element as mandatory (marked as "M" in the according "Content of Function <function> at Actor <actor>" table) to fulfill a specific Use Case Scenario, the device SHALL fully support and transmit this data in its full notify or full reply messages (for messages using Restricted Function Exchange (RFE): see below) and it SHALL interpret the Element in each message received from the communication partner (write messages). The server SHALL reject any received write command, that changes or adds Elements that are not writeable (except for Primary and Sub-Identifiers, see below).
- For a client: If a Use Case specification defines an Element as mandatory (marked as "M" in the according "Content of Specialization <specialization> at Actor <actor>" table) to fulfill a specific Use Case Scenario, the device SHALL fully support and transmit this data in its full write messages*[1] (for messages using Restricted Function Exchange (RFE): see below) and it SHALL interpret the Element in each message received from the communication partner (notify and reply messages).

*[1]: In cases where not all Elements are writeable by a client (which is the case in many Use Cases), the client SHALL use RFE for the write command. Otherwise, the full write command would be rejected by the recipient.

For all messages, the following rule holds valid: All Primary- and (available) Sub-Identifiers SHALL be included in the message, regardless of being writeable or not.

Note: Within RFE messages, even mandatory Elements MAY be skipped as only changed Elements must be transmitted (this holds valid for notify and write messages; for reply messages, it depends on the filter set by the client-side).

3.2 Subscription and data retrieval management

This section clarifies the requirements and best practices for establishing data monitoring via subscriptions as defined in each "Pre-scenario communication" of a Use Case (in the sub-sections of section 3.4 of each Use Case) and the detailed pre-scenario communication description for subscriptions (section 3.3.4 of each Use Case).

3.2.1 Definition of "relevant" Features

The Scenario specific sub-section withing section 3.4 of each Use Case state that actors SHALL create a subscription for each server Feature that is "relevant" for the corresponding Actor.

- Clarification: A Feature is considered relevant if it is explicitly listed in the "Initial Scenario communication" or "Runtime Scenario communication" sequences of the respective Scenario.
- Requirement: Implementers must cross-reference the pre-scenario subscription phase with the subsequent runtime chapters. Any data point that the Actor expects to receive via a notify or intends to write to SHALL be covered by a subscription request during the discovery/setup phase.

3.2.2 Priority of subscriptions over polling

There is an apparent contradiction between the mandatory support for subscriptions ("SHALL support subscription") and the description of fallback behavior ("SHOULD read data periodically if subscription fails").

- Clarification: The mandatory "SHALL" requirement for subscription support is the primary rule for EEBUS compliance. The fallback "polling" logic described in section 3.3.4 is intended strictly for exceptional error handling (e.g., reaching temporary resource limits) or interoperability with legacy/constrained devices.
- Requirement: A device SHALL demonstrate that it successfully establishes and maintains subscriptions. Polling SHALL NOT be used as a primary data retrieval strategy if the server indicates support for the corresponding subscription.

3.2.3 Prohibition of redundant polling

A common anti-pattern is "parallel polling," where a client continues to send read requests even though a successful subscription has been established and notify messages are being received.

- Best practice: If a subscription is active and the client is receiving notify messages, the client SHOULD NOT perform additional polling (periodic read requests) for the same data points.
- Reasoning: Redundant polling creates unnecessary network congestion, increases the processing load on both the client and server and provides no additional data integrity.

3.3 Binding partner clarifications (general information)

Use Cases that provide writeable data points specify in their Scenario specific sections within the "Technical SPINE solution": "Only one binding partner is allowed to write the data specified in this Scenario.". This does not mean that only one binding partner is permitted in total for the device. It just means that the Server Actor SHALL NOT permit more than one of its binding partners to modify those data points that are changeable by the Client Actor.

With the upcoming SPINE V1.4.0, at maximum one binding partner will be permitted per Feature. This means that if one communication partner (device "B") requested a binding on a Feature on device "C" and that binding was accepted by device "C", no further binding request on this Feature will be permitted by "C".

More details may be given in the Use Case specific implementation guides.

Annex A - Classification of Use Case Actors as Client or Server Roles

This annex provides a reference matrix for all EEBUS Use Cases. In accordance with the functional definitions established in section 2.1 of this document, the high-level Actor roles of each Use Case are explicitly categorized into either Client Actor or Server Actor roles.

Use Case name	Short name	Client Actor(s)	Server Actor(s)
Configuration of DHW System Function	CDSF	Configuration Appliance	DHW Circuit
Configuration of DHW Temperature	CDT	Configuration Appliance	DHW Circuit
Configuration of Room Cooling System Function	CRCSF	Configuration Appliance	HVAC Room
Configuration of Room Cooling Temperature	CRCT	Configuration Appliance	HVAC Room
Configuration of Room Heating System Function	CRHSF	Configuration Appliance	HVAC Room
Configuration of Room Heating Temperature	CRHT	Configuration Appliance	HVAC Room
Control of Battery	COB	CEM	Inverter
Coordinated EV Charging	CEVC	Energy Guard Energy Broker	EV
Dynamic Bidirectional EV Charging	DBEVC	CEM	EV
EV Charging Electricity Measurement	EVCEM	CEM	EV
EV Charging Summary	EVCS	Energy Broker	EVSE
EV Commissioning and Configuration	EVCC	CEM	EV
EV Peak Shaving	EVPS	Energy Manager	Charging Station Manager
EV State of Charge	EVSOC	Monitoring Appliance	EV
EVSE Commissioning and Configuration	EVSECC	CEM	EVSE
Extra Power Request	EPRQ	Transmission Broker	CEM
Flexible Load	FLOA	CEM	Energy Consumer
Flexible Start for White Goods	FSWG	CEM	Smart Appliance
Incentive-Table based power consumption management	ITPCM	CEM	Energy Consumer
Limitation of Power Consumption	LPC	Energy Guard	Controllable System
Limitation of Power Production	LPP	Energy Guard	Controllable System
Monitoring and Control of Smart Grid Ready Conditions	MCSGRC	CEM	Heat Pump
Monitoring of Battery	MOB	Monitoring Appliance	Battery
Monitoring of DHW System Function	MDSF	Monitoring Appliance	DHW Circuit
Monitoring of DHW Temperature	MDT	Monitoring Appliance	DHW Circuit

Monitoring of Grid Connection Point	MGCP	Monitoring Appliance	Grid Connection Point
Monitoring of Inverter	MOI	Monitoring Appliance	Inverter
Monitoring of Outdoor Temperature	MOT	Monitoring Appliance	Outdoor Temperature Sensor
Monitoring of Power Consumption	MPC	Monitoring Appliance	Monitored Unit
Monitoring of PV String	MPS	Monitoring Appliance	PV String
Monitoring of Room Cooling System Function	MRCSF	Monitoring Appliance	HVAC Room
Monitoring of Room Heating System Function	MRHSF	Monitoring Appliance	HVAC Room
Monitoring of Room Temperature	MRT	Monitoring Appliance	HVAC Room
Node Identification	NID	Visualization Appliance	Identifiable Node
Optimization of Self Consumption by Heat Pump Compressor Flexibility	OHPCF	CEM	Compressor
Optimization of Self-Consumption During EV Charging	OSCEV	CEM	EV
Overload Protection by EV Charging Current Curtailment	OPEV	Energy Guard	EV
Power Demand Forecast	PODF	Energy Broker Transmission Broker	CEM
Power Envelope	POEN	Transmission Broker	CEM
Time of Use Tariff	TOUT	Energy Broker Transmission Broker	CEM
Visualization of Aggregated Battery Data	VABD	Visualization Appliance	Battery System
Visualization of Aggregated Photovoltaic Data	VAPD	Visualization Appliance	PV System
Visualization of Heating Area Name	VHAN	Visualization Appliance	Heating Circuit, Heating Zone, HVAC Room
Visualization of Electrical Power Consumption of the Heat Pump Compressor	VPCHPC	CEM	Heat Pump Compressor

256 *Table 1: Classification of Use Case Actors as Client or Server Roles*

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